

Experimental Inter-Report Scatter Bounds Machine-Learning Accuracy for TADF Singlet–Triplet Gaps—Structure and Semi-Empirical Features Reach the Same Limit

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The singlet–triplet gap ΔE_{ST} is the screening parameter for thermally activated delayed fluorescence (TADF), yet independent experimental reports of the same compound scatter by 0.072 eV RMS, and the per-molecule median regressed on is known only to 0.049 eV, about half of any model's error. This noise floor, not model capacity, bounds every predictor. Against that ceiling, two-dimensional structure predicts as accurately as a semi-empirical excited-state calculation: on 231 donor–acceptor molecules (212 Bemis–Murcko scaffolds, scaffold cross-validation), random forests from Morgan fingerprints (MAE 0.091 eV) and RDKit descriptors (0.100 eV) match natural-transition-orbital descriptors from a GFN2-xTB/sTDA excited state (0.096 eV) to within 0.017 eV on paired folds, so the excited-state step buys no accuracy. Two further results are specific to this corpus. First, under a source-paper split the accuracy advantage over a mean-value baseline disappears, but the excited-state descriptors keep the rank correlation the fingerprints lose (paired $\Delta\rho = 0.19$ on a single split, 95% CI narrowly excluding zero): they order candidates across laboratories without predicting the gap. Second, the leakage that inflates many computed-gap models is learner-dependent — given the S_1 and T_1 energies as features, a linear model returns $\Delta E_{\text{ST}} \equiv E_{S_1} - E_{T_1}$ exactly ($R^2 = 1.000$) while an axis-aligned forest on the same features recovers only $R^2 = 0.59$ and clears cross-validation unremarked. Progress past the ≈ 0.1 eV ceiling needs a curated, solvent-controlled, single-source experimental benchmark rather than a better model. Data, models, and code are available (Zenodo DOI: 10.5281/zenodo.17436069).

1 Introduction

1.1 Predicting the singlet–triplet gap of TADF emitters

Thermally activated delayed fluorescence (TADF) emitters—organic donor–acceptor molecules that harvest both singlet and triplet excitons for near-unity internal quantum efficiency¹—are the heavy-metal-free route to tunable, efficient light sources for displays and for human-centric and horticultural lighting^{2,3}. An efficient emitter has to meet several requirements at once: a small singlet–triplet gap ($\Delta E_{\text{ST}} < 0.2$ eV), fast reverse intersystem crossing ($k_{\text{RISC}} > 1 \times 10^6 \text{ s}^{-1}$), and emission matched to the target band. This paper addresses only the first. The rate and device-efficiency requirements need spin–orbit-coupling and device-level calculations beyond the cheap structural data used here, and we make no quantitative claim about them (section 3.8).

1.2 The computational bottleneck and the role of machine learning

Designing such emitters means searching a large space of donor–acceptor combinations for a small ΔE_{ST} . The direct route—time-dependent DFT on excited states—scales steeply with system size⁴, so exhaustive screening of more than a few hundred candidates is impractical. Machine learning offers a cheaper surrogate^{5,6}, and several groups have predicted TADF gaps from molecular descriptors^{7,8}. Those low reported errors, however, are obtained against *computed* (TD-DFT or higher) reference gaps; predicting the *experimental* gap is markedly harder, to the point that one study abandoned direct regression for classification⁹ (section 3.7).

Two problems recur in this literature, and we confront both. First, accuracy is often quoted against a single random train–test split, which rewards interpolation within familiar scaffolds and overstates how a model behaves on new chemistry. Second, and more subtle, models are sometimes trained to reproduce a *computed* gap from a feature set that already contains the constituent excitation energies; because $\Delta E_{\text{ST}} \equiv E_{S_1} - E_{T_1}$, this reduces to subtraction and the reported accuracy reflects arithmetic rather than learning.

Our earlier work set the stage: Article 1¹⁰ benchmarked in JCIM GFN2-xTB/sTDA against 747 experimental molecules (MAE = 0.162 eV), xTB geometry relaxations, and scaffold cross-validation, and Article 2¹¹ extracted structure–property design rules from that set. Here we ask a sharper question: what sets the ceiling on predicting the *measured* ΔE_{ST} from molecular structure, and what would it take to raise it?

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1.3 Our contributions

We do not claim a new high-accuracy predictor. The primary finding is that 2D structure alone predicts the measured singlet-triplet gap as well as semi-empirical excited-state features, because the experimental label, not the model, is the binding constraint. Two further results are specific to this task and, to our knowledge, are not reported elsewhere: the rank order that 2D structure encodes and the rank order that only an excited-state calculation encodes come apart under a change of source laboratory, and the feature-target leakage that inflates computed-gap models is visible to some learners and hidden from others. Four points follow.

Structure alone predicts as well as the semi-empirical excited state. On 231 molecules across 212 Bemis-Murcko scaffolds, random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—match descriptors derived from a semi-empirical excited-state calculation (natural transition orbital, NTO, spatial features, 0.096 eV). The result holds across scaffold, cluster and random cross-validation and against a label-permutation null, so it is not a splitting artefact. On accuracy the excited-state step buys nothing: the two representations agree to within 0.017 eV on paired folds. On ranking under laboratory transfer, however, the two representations separate. Under a DOI-grouped split—holding out whole source publications rather than individual molecules—the NTO model retains Spearman $\rho = 0.35$ while Morgan drops to 0.16 ($\Delta\rho = 0.19$, 95 % CI 0.04 to 0.34, excluding zero only narrowly and on a single split, with resampling over molecules rather than the 173 source papers). The excited-state features do not improve accuracy on the gap; they preserve rank order that transfers across source laboratories—a dissociation we have not seen reported for this property. SHAP analysis ties the NTO signal to hole-electron overlap and charge transfer, giving an interpretable read on what structure encodes. Unlike a physics-guided model trained on NTO descriptors against a *computed* reference gap¹², we test the same class of descriptor against the *measured* gap and find the physical-guidance step buys no accuracy there either.

The experimental noise floor, quantified. Why does accuracy plateau near 0.1 eV regardless of descriptor set? Independent experimental reports for the same compound scatter by 0.072 eV RMS (the worst molecule reaching a standard deviation of 0.56 eV), and the median gap the model is trained on is itself known only to 0.049 eV—about half the model error. That ΔE_{ST} is reported inconsistently has been shown for individual emitters¹³; what we add is the size of that inconsistency over 231 molecules and 173 source papers and its direct consequence for prediction—to our knowledge the first inter-laboratory variability budget for the TADF singlet-triplet gap. The label bounds what any predictor can reach, consistent with the broader finding that many machine-learning benchmarks in drug-, materials- and molecular-discovery are limited by dataset noise rather than model capacity¹⁴—applying their $R^2_{\text{max}} = 1 - \sigma^2_{\text{noise}}/\sigma_y^2$ framework to our label distribution gives $R^2_{\text{max}} \approx 0.71\text{--}0.91$ depending on which noise floor applies (section 3.4)—and with the aleatoric/epistemic distinction under which label noise is an irreducible floor no model can cross¹⁵. Computed references face a separate limit: B3LYP and CAM-B3LYP gaps differ by up to 0.58 eV, so a gap computed at one level of theory does not transfer to another; that bound constrains the meaning of a computed reference, not the error of a model fitted to measured labels. The 0.1 eV plateau is not a model-capacity failure: neither a higher-capacity tabular learner (gradient boosting, MLP, SVR) nor a frozen pretrained molecular-transformer embedding beats the random forest (the latter reaches only MAE 0.109 eV to 0.113 eV), and the result places itself honestly at $R^2 \approx 0.25$, a real but limited signal.

Feature-target leakage, and why it survives review. A feature that is a proxy for the target—here, an exact one—is a long-recognised cause of over-optimistic ML results^{16–18}, and the apparent severity of such a leak is known to depend on the model class. What we add is the controlled measurement for this specific case. Regressing ΔE_{ST} on features containing the S_1 and T_1 energies is circular ($\Delta E_{\text{ST}} \equiv E_{S_1} - E_{T_1}$): under the scaffold cross-validation used throughout, a linear model given those energies recovers the gap exactly ($R^2 = 1.000$, MAE = 0) against $R^2 = 0.36$ without them. An axis-aligned random forest does not recover that subtraction—its splits approximate a difference of two continuous inputs rather than express it, a familiar limitation of tree ensembles—so it reaches only $R^2 = 0.59$ on the identical leaked features, a score that does not look anomalous, which is how a tree-based study can carry this defect through cross-validation unremarked. The lesson transfers to any computational-chemistry ML pipeline that regresses on computed energy components.

Modest triage utility, and what would be needed to push past the ceiling. As a pre-screen, the model enriches low-gap candidates (1.2 to 1.4 $\times$ over random selection) on held-out labelled data, though a bootstrap confidence interval shows this only holds at cutoffs of 50 molecules or more—at the ten-to-twenty-candidate scale a laboratory would actually shortlist, the enrichment is not distinguishable from chance. On the filtered virtual library the conformal prediction intervals are too wide to support molecule-level selection; no ranked shortlist is deposited. ΔE_{ST} is necessary but not sufficient for TADF efficiency, and we make no device-level performance claims. Progress past the ≈ 0.1 eV ceiling requires not a better model but a better dataset: a curated, single-source, solvent-controlled experimental benchmark of sufficient size to push the label precision below the present 0.049 eV floor. This study provides the quantitative justification for that investment.

This study builds on our earlier work—**Article 1**¹⁰ (xTB/sTDA validation) and **Article 2**¹¹ (design rules)—by measuring, rigorously, what such cheap data can and cannot support. All data, trained models, and code are openly available (Zenodo DOI: 10.5281/zenodo.17436069), enabling reproduction and extension of the benchmark.

2 Methods

2.1 Experimental target and dataset

The prediction target is the *experimental* singlet-triplet gap ΔE_{ST} . We extracted literature-reported values from a curated corpus of TADF publications and matched them, by compound name, to molecules in our descriptor set, retaining values in the physical range -0.3 eV to 1.5 eV. Where a compound carried multiple independent reports we used the median and recorded the inter-report spread; a single report sits 0.072 eV from the median (RMS, weighting each molecule equally; the report-pooled value is

0.0905 eV), and the median itself carries an RMS standard error of 0.049 eV, which is the precision of the target actually regressed on (section 3.4). The final benchmark comprises 231 donor-acceptor molecules spanning 212 distinct Bemis-Murcko scaffolds¹⁹.

The corpus does not record measurement medium (solvent, host) for individual values, so the per-molecule median is a consensus over uncontrolled conditions rather than a single, solvent-specific state. Because environmental electrostatics and conformational disorder are recognised as primary obstacles to accurate TADF modelling,²⁰ this consensus is best read as the central tendency of experimentally accessible S_1 - T_1 gaps; it trades physical specificity for reduced variance. For the 102 molecules with repeat reports, a single random report deviates from this median by 0.072 eV on average—comparable to the model error itself (table S3). The median itself is known more precisely, to 0.049 eV RMS, and that is the figure relevant to attainable accuracy.

We deliberately do *not* train on the semi-empirical gap $E_{S_1} - E_{T_1}$: because this quantity is, by construction, a difference of two input features, regressing onto it measures arithmetic rather than learning (section 3.1).

2.2 Cheap electronic-structure inputs and physics-informed descriptors

Excited-state quantities used to build descriptors were obtained at the semi-empirical level: GFN2-xTB²¹ geometries (`xtb` version 6.7.1) followed by sTDA²² for the lowest singlet and triplet states. From the natural transition orbitals (NTOs)²³ of each excitation we computed spatial overlap descriptors that encode exchange-integral physics:

$$S_{\text{he}} = \int \phi_{\text{hole}}(\mathbf{r}) \phi_{\text{electron}}(\mathbf{r}) d\mathbf{r}, \quad (1)$$

$$\Delta\lambda_{\text{A}} = |\lambda_{\text{A}}(S_1) - \lambda_{\text{A}}(T_1)|, \quad (2)$$

$$\text{Char}_{\text{diff}}^2 = \left[\sum_i |q_i(S_1) - q_i(T_1)| \right]^2, \quad (3)$$

$$\Delta S_{\text{he}} = |S_{\text{he}}(S_1) - S_{\text{he}}(T_1)|. \quad (4)$$

S_{he} quantifies hole-electron overlap (related to the exchange integral), $\Delta\lambda_{\text{A}}$ acceptor-participation differences, $\text{Char}_{\text{diff}}^2$ charge-transfer character changes, and ΔS_{he} the singlet-triplet overlap difference. The full set of 35 NTO descriptors (the S_1/T_1 energy scalars excluded) is listed in the Supporting Information; these descriptors require the sTDA excited-state calculation, even though they omit the energies themselves.

2.3 Feature sets and machine-learning models

We compared three feature representations on the same molecules. Two require no quantum chemistry, being computed from the 2D structure alone: (i) RDKit physicochemical descriptors²⁴ and (ii) Morgan fingerprints (radius 2, 2048 bits). The third, (iii) the NTO spatial descriptors above, is derived from the sTDA excited state but excludes the energy scalars. A reference set that additionally includes the sTDA S_1/T_1 energies was also evaluated. We used random-forest regressors²⁵ (400 trees, scikit-learn²⁶), which outperformed support-vector regression, ridge regression and a mean-value baseline on this task.

Evaluation. All performance figures use Bemis-Murcko scaffold `GroupKFold` ($k=5$). Because 92.9% of scaffolds are singletons, this split is numerically close to a random split; we therefore also report a source-paper (DOI-grouped) split, which holds out a synthetic series and whatever measurement practice is shared within a laboratory, and is the harder test of transfer (section 3.2). The corpus records no solvent, host or temperature per value, so the source paper is a *proxy* for protocol, not a measured one. We benchmark against a mean-value predictor, the raw sTDA gap, and a range-separated TD-DFT reference gap (section 3.2), and report 95% confidence intervals from 2000 bootstrap resamples of the out-of-fold predictions.

TD-DFT reference gap. For a higher-level reference we computed vertical ΔE_{ST} for all 231 molecules at wB97X-D4/def2-SVP with RIJCOSX and the Tamm-Dancoff approximation (ORCA 6²⁷; `nroots` 8, triplets included; GFN2-xTB ground-state geometry). All 231/231 jobs converged (76.2 h elapsed on 8 processes, so 609 core-hours; median 16.1 min per molecule, slowest 2.1 h). These are measured wall times from `results/tddft_reference_231.csv`, not a scaling extrapolation. The raw gap, an in-fold linearly calibrated gap, and the gap as an added descriptor in the random forest model are evaluated against the mean-value baseline on the same scaffold and DOI cross-validation folds (section 3.2; table S15).

Interpretability and reproducibility. Feature importance was quantified with exact TreeSHAP²⁸ on the fitted forest. All scripts (`code/finalize_model.py`, `code/rebuild_structural_model.py`) regenerate every reported number and fig. 1.

Bootstrap confidence intervals and power analysis. Point estimates of the rank correlation are complemented by 5000-replicate percentile bootstrap confidence intervals on the out-of-fold predictions (`code/r2_ci_power_analysis.py`). For the headline model the Spearman $\rho = 0.36$ ($n = 231$) has a 95% CI of 0.24 to 0.47 ($p = 2.6 \times 10^{-8}$) and the Pearson $r = 0.50$ a 95% CI of 0.26 to 0.68; both exclude zero, establishing statistically significant association despite the wide R^2 interval. A power analysis for the correlation test (Fisher z transform, two-sided, $\alpha = 0.05$) shows that detecting $\rho = 0.36$ at 80% power requires only $n \approx 60$ (90% power: $n \approx 79$); at $n = 231$ the achieved power is ≈ 1.0 . This bears on the ranking question only: it establishes that the ρ estimate is not sample-size-limited, and it says nothing about the width of the R^2 interval, which is a different statistic and is discussed on its own terms in section 3.2. We report Spearman ρ as the primary metric for triage utility because triage is a ranking task, not because it is the more favourable number; the full set of metrics and intervals is collected in table S6.

2.4 Reference-level benchmark

To bound the meaning of any model accuracy, we compared the computed reference across functionals for 7 representative emitters in gas and toluene, contrasting B3LYP/6-31G(d) with CAM-B3LYP/def2-TZVP (ORCA 6.1.1²⁷). The two levels differ in ΔE_{ST}

by a mean absolute 0.30 eV (up to 0.58 eV, with cases of reversed relative magnitude); full values are tabulated in table S10. This functional dependence bounds what a *computed* gap can mean; it is reported separately from the label-precision term that applies to the present regression, and the two are not combined (section 3.4). The same CAM-B3LYP/def2-TZVP calculations additionally yielded excited-state-optimised (adiabatic) gaps and the two lowest triplets (T_1 , T_2) for 14 emitters, used to bound the vertical approximation (section 3.4) and the triplet-manifold scope (section 3.8); see SI.

2.5 Substructure-filtered extended library

To gauge the scale of chemical space beyond the annotated set, we screened the PubChem CID-SMILES collection for compounds carrying *both* a donor and an acceptor motif, retaining 22 194 structures. The donor patterns were carbazole, phenothiazine, phenoxazine, acridine, diphenylamine, triphenylamine and dimethylaniline; the acceptor patterns triazine, benzonitrile, pyridine, pyrimidine, quinoxaline, benzothiadiazole, naphthoquinone and a bare cyano group. Additional constraints were molecular weight < 900 Da, ≥ 2 aromatic rings and ≥ 1 N/O/S heteroatom (`code/build_shortlist.py` and the screening script in the deposit). This is a substructure filter over an existing database, not a combinatorial enumeration of fragment pairs, and the distinction matters for how the set should be read: every member satisfies the donor-acceptor pattern test, but the matches are dominated by the weaker motifs (dimethylaniline in 85 % and pyridine in 59 % of the 100 lowest-predicted structures, against carbazole in 2 and triazine in 3). The set also inherits the composition of the source database rather than of a designed emitter series: 97 % of those structures carry a halogen and 53 % a stereocentre, and enantiomer pairs appear as separate entries with identical predicted gaps, which a 2D fingerprint cannot distinguish. These were ranked by a rule-based TADF score

$$\text{TADF score} = 0.35 f_{\text{MW}} + 0.25 f_{\text{LogP}} + 0.20 f_{\text{rings}} + 0.20 f_{\text{hetero}}, \quad (5)$$

with each $f_i \in [0, 1]$ a normalised feature score based on established design heuristics^{1,29,30}. The score is a coarse synthesis-planning filter, *not* a calibrated ΔE_{ST} predictor; the filtered molecules are not quantum-validated and are reported as leads only. The random-forest model trained on the labelled benchmark is applied to this set only to obtain split-conformal prediction intervals (section 3.6); no candidate is put forward on the strength of its predicted gap, and the descriptor pipeline and cross-validation protocol are those of the labelled benchmark throughout.

2.6 Data and code availability

The experimental corpus, descriptor tables, trained models, and all analysis and figure-generation scripts are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under a CC-BY-4.0 licence, so that every number and figure in this work can be reproduced and the benchmark extended.

3 Results and Discussion

3.1 Why we target the measured gap: ruling out a circular benchmark

A tempting shortcut is to regress the singlet-triplet gap ΔE_{ST} on a feature set that already contains the S_1 and T_1 energies. Because $\Delta E_{\text{ST}} \equiv E_{S_1} - E_{T_1}$, such a model only subtracts two inputs: we verified this directly. Regressing the computed gap on the NTO descriptors *plus* the two energy scalars, under the same scaffold cross-validation used throughout, a linear model recovers the target exactly (MAE = 0, $R^2 = 1.000$) while the identical features without the energies reach only $R^2 = 0.36$ (table S4). The recovered quantity is arithmetic, not learning. How severe the leak looks depends on the learner: a random forest, being axis-aligned, approximates a difference of two continuous inputs through its splits rather than expressing it, and reaches only $R^2 = 0.59$ on the same leaked features (a joint effect of that inductive bias, tree depth and the 231-molecule sample)—which is how a tree-based study can carry this defect without its cross-validated score appearing anomalous. We therefore discard the computed gap as a target and benchmark on the only non-trivial quantity: the *experimentally measured* ΔE_{ST} . This risk is not particular to TADF: any regression target that is an algebraic combination of features already in the descriptor set is vulnerable to the same leak, and how visible it is under cross-validation depends on the learner, not only on the target.

3.2 Structure-based models predict experimental gaps with modest, honestly-bounded accuracy

We assembled 231 donor-acceptor molecules for which an experimental ΔE_{ST} could be matched to a structure, and evaluated random-forest regressors under Bemis-Murcko scaffold cross-validation (212 distinct scaffolds), so that no scaffold appears in both training and test. Because 92.9 % of scaffolds are singletons, scaffold and random splits produce numerically similar folds (test-train nearest-neighbour Tanimoto differs by only 0.022). The scaffold split guarantees zero shared scaffolds per fold (versus 3 to 8 under a random split) but task difficulty barely changes; the harder test of transfer is the source-paper split reported below. That the model learns real signal rather than scaffold-group structure is confirmed by a label-permutation test: the true scaffold-CV MAE (0.096 eV) lies far below the null from 1 000 target shuffles (mean 0.120 eV, $p = 0.001$; table S9). The headline accuracy is also stable across validation schemes: replacing the singleton scaffolds with genuine multi-molecule Butina clusters (down to 70 groups, largest 38 molecules) leaves the MAE at 0.099 eV to 0.104 eV, between scaffold-CV (0.096 eV) and random-CV (0.106 eV) (table S17). The three feature sets perform within 0.009 eV of one another (table 1); for the fingerprint and excited-state models, which we compare head to head below, that closeness is backed by a paired bound of 0.017 eV. Two are computed from the 2D structure alone, with no quantum chemistry: Morgan fingerprints (MAE = 0.091 eV) and RDKit physicochemical descriptors (0.100 eV). The third, NTO spatial descriptors, is derived from a semi-empirical sTDA excited-state calculation but excludes the S_1/T_1 energy scalars; it reaches MAE = 0.096 eV (95 % CI 0.082 to 0.109), $R^2 = 0.25$ (CI 0.03 to 0.45), Spearman $\rho = 0.36$. That the structure-only sets match the NTO set is the central methodological point: the excited-state calculation buys no accuracy, and adding back the S_1/T_1 energies does not help either (table 1). Equivalently, the spatial NTO features carry no information beyond what the energy difference

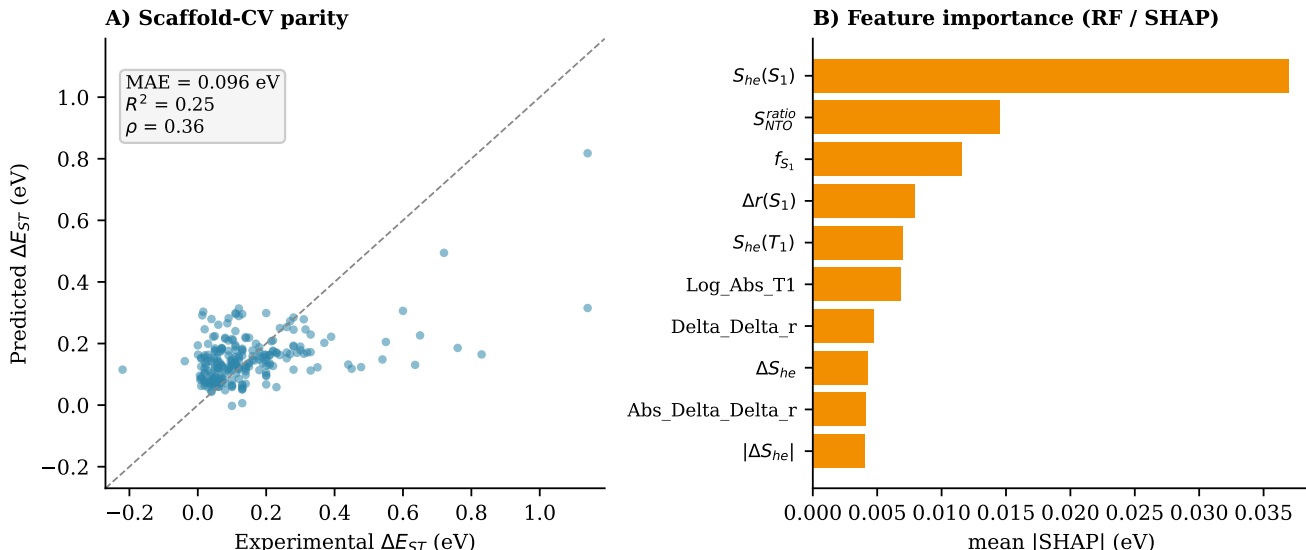


Figure 1 Structure-based prediction of experimental ΔE_{ST} . (A) Scaffold cross-validated parity for the random forest on NTO spatial descriptors (231 molecules, 212 Bemis–Murcko scaffolds): MAE = 0.096 eV, $R^2 = 0.25$, Spearman $\rho = 0.36$. Models built from the 2D structure alone (Morgan, RDKit) perform equivalently (table 1). The model regresses high-gap outliers toward the mean, the expected behaviour when the training distribution is dominated by low-gap emitters. (B) SHAP importances: hole–electron overlap S_{he} , NTO overlap ratio, oscillator strength f_{S_1} and charge-transfer distance Δr dominate, consistent with exchange-integral physics.

already provides: the MAE of the *Energies + NTO* set (0.097 eV) matches the spatial-only NTO set (0.096 eV) to 0.001 eV. The natural comparator is the mean-value baseline (MAE = 0.107 eV), over which every model improves by only about 10%; the large apparent gain over the raw sTDA gap (0.25 eV) reflects that the semi-empirical gap is badly biased, not that the model is accurate. The R^2 confidence intervals are wide and, for the structure-only sets, include negative values (RDKit -0.11 to 0.51 ; Morgan -0.19 to 0.53): at 95% confidence we cannot exclude that these models are no better than predicting the mean. The signal is real but weak.

A higher level of theory does not change this picture. A range-separated TD-DFT gap (wB97X-D4/def2-SVP, Tamm–Dancoff, 609 core-hours) overestimates the measured gap by a systematic 0.65 eV and, used directly, is the least accurate predictor in table 1 (MAE 0.654 eV, $R^2 = -16.3$). Its correlation with experiment is higher than the semi-empirical gap’s (Pearson r from 0.32 to 0.43), and an in-fold linear calibration that removes the offset brings it to MAE 0.099 eV—level with the mean-value baseline, still short of the structure-only forest. Added as a single feature it lowers the Morgan model from 0.091 to 0.087 eV (paired 95% CI 0.0004 to 0.0071, Wilcoxon $p = 0.016$): a real but 4% effect for 609 core-hours (table S15). The excited-state step, semi-empirical or TD-DFT, does not move the structure-only result.

Holding out whole source papers rather than scaffolds is a stricter test: it removes a synthetic series and a laboratory’s shared measurement practice together — a proxy for protocol, since the corpus records no measurement conditions per value. Under a DOI-grouped GroupKFold (173 papers, 231 molecules) the Morgan model reaches MAE 0.102 eV ($R^2 = 0.18$), and its paired advantage over the mean-value baseline falls to 0.005 eV with a 95% CI that crosses zero. On *accuracy*, then, the model does not demonstrably transfer to an unseen source laboratory; part of the scaffold-CV signal is within-paper regularity.

Ranking behaves differently, and this is the one place the excited-state features separate from the fingerprints. Across the same DOI folds the NTO model holds Spearman $\rho = 0.35$ while Morgan drops to 0.16 (MAE 0.099 eV against 0.102 eV); the paired difference $\Delta\rho = 0.19$ has a 95% CI of 0.04 to 0.34 and excludes zero, though only just; the resampling is over molecules rather than over the 173 source papers, so a group-level interval would be wider, and this is a single comparison on a single split rather than a general result. Since Spearman ρ is the metric we report as primary for triage (section 2), the semi-empirical excited-state step earns its cost for ordering candidates across laboratories even though it earns nothing for the value of the gap. One observation bounds the pessimistic reading, with a caveat that must travel with it: 58% of repeatedly-measured molecules have zero inter-report spread, and on the 102-molecule multi-report subset the model reaches the study’s lowest MAE, 0.080 eV. That subset is also an easier target: a mean-value predictor restricted to it scores 0.089 eV, so the model’s advantage there is 0.009 eV with a 95% CI spanning zero. The low absolute error reflects the narrower spread of those labels more than any gain in prediction.

For the NTO variant, SHAP (SHapley Additive exPlanations) analysis (fig. 1B) identifies hole–electron overlap S_{he} , the NTO overlap ratio, oscillator strength f_{S_1} and the charge-transfer distance Δr as the dominant predictors—spatial quantities that encode the exchange integral governing ΔE_{ST} . This ranking is stable rather than an artefact of one fit: across 200 bootstrap refits the leading descriptor $S_{he}(S_1)$ stays in the top five 99.5% of the time, and on average 3.7 of the five canonical descriptors recur (table S8). Restricting the attribution to the reliable-prediction subset—the 152 molecules whose out-of-fold residual is below 0.10 eV—leaves $S_{he}(S_1)$ ranked first in 99.5% of refits, so the leading attribution does not rest on the poorly-predicted tail. An independent, model-agnostic permutation-importance analysis ranks the same descriptor, $S_{he}(S_1)$, first, so the leading attribution is corroborated by

Table 1 Prediction of experimental ΔE_{ST} under scaffold cross-validation (random forest, 231 molecules); R^2 with 95 % bootstrap CI. The 2D-structure-only sets (no quantum chemistry) match the NTO set, which requires an sTDA excited-state run; none approaches the precision a quantitative photophysical method would require, and the structure-only R^2 intervals include zero. The range-separated TD-DFT reference gap (wB97X-D4/def2-SVP, Tamm–Dancoff; 609 core-hours) is listed raw, after an in-fold linear calibration, and as an added feature (table S15).

Feature set	MAE (eV)	R^2 (95% CI)	Spearman ρ
Morgan FP (2D structure only)	0.091	0.27 (−0.19, 0.53)	0.31
RDKit desc. (2D structure only)	0.100	0.31 (−0.11, 0.51)	0.26
NTO spatial (needs sTDA)	0.096	0.25 (0.03, 0.45)	0.36
Energies + NTO (reference)	0.097	0.24	0.34
Mean-value baseline	0.107	0.00	—
Raw sTDA gap	0.251	−2.22	0.24
<i>Physics reference gaps (higher level of theory)</i>			
Raw TD-DFT gap (wB97X-D4/def2-SVP)	0.654	−16.3	0.38
Calibrated TD-DFT gap (in-fold)	0.099	0.17	0.37
Morgan FP + TD-DFT gap	0.087	0.31	0.37

two methods (the lower-ranked tail is not; SI section S6.3). Given the model’s modest ranking power, we read only the dominant attribution as a mechanistic hypothesis for experimental test rather than a validated causal claim; the structure-only fingerprint and descriptor models reach the same accuracy without any excited-state calculation.

3.3 Accuracy is equivalent; ranking under laboratory transfer is not

Do quantum-mechanical features beat pure structural descriptors when the target is an experimental property? For accuracy, no, and we can put a bound on the equivalence. Paired on identical folds, the NTO model (MAE = 0.096 eV) and the Morgan model (0.091 eV) differ by $\Delta\text{MAE} = -0.004$ eV (95 % CI −0.017 to 0.009, Wilcoxon $p = 0.66$). The two feature sets are therefore equivalent to within 0.017 eV—an equivalence margin roughly a third of the 0.049 eV precision of the label itself, not merely a failure to reject. Both beat the mean-value baseline by 10 % to 15 % MAE (paired CI excludes zero), and the excited-state step adds nothing to that margin.

The equivalence is a statement about *accuracy*, and it does not extend to *ranking* under protocol transfer. Under the source-paper split (section 3.2) the NTO model retains Spearman $\rho = 0.35$ while the Morgan model falls to 0.16; the paired difference is $\Delta\rho = 0.19$ (95 % CI 0.04 to 0.34, excluding zero). On the labelled benchmark the same asymmetry appears in triage: at the top 20 % the NTO model enriches low-gap emitters 1.6-fold against 1.4-fold for Morgan. The excited-state calculation does not buy accuracy on the gap; it buys ordering that survives a change of source laboratory, which is what a triage filter is actually asked to do. Recent theoretical work by Salvi et al.³¹ on electronic correlation effects explains why semi-empirical excited-state features may not outperform 2D features against experimental gaps: at the semi-empirical level, the charge-transfer excitation energies are not quantitatively accurate enough to provide additional discrimination beyond what 2D topology already encodes. The two representations differ in what they can be read for, though not in a way the attribution totals establish on their own. The ten highest-attribution NTO descriptors carry 79 % of that model’s SHAP weight against 30 % for the ten highest-attribution Morgan bits (fig. S2); those totals are not comparable, because ten descriptors are 29 % of the 35-dimensional NTO set and ten bits are 0.5 % of a 2048-bit fingerprint, so the concentration follows from dimensionality. What does not follow from dimensionality is that the NTO features are *named* donor–acceptor overlap and charge-transfer quantities, whereas Morgan bits are anonymous hashed substructures. The case for the physics-informed representation is interpretability of the variables themselves, not a larger share of attribution.

3.4 What limits achievable accuracy

Accuracy near 0.1 eV is limited by the data rather than by the model, and the term that applies is the quality of the label. Independent experimental reports for the same compound disagree substantially: a single randomly chosen report sits 0.072 eV from its molecule’s median (RMS with each molecule weighted equally), the disagreement exceeds the model MAE for 19.6 % of molecules, and the worst molecule reaches a standard deviation of 0.56 eV; the eight independent values for 4CzIPN alone span 0.01 eV to 0.48 eV (table S3). That ΔE_{ST} is reported inconsistently has been shown for individual emitters¹³; quantified across 231 molecules and 173 source papers it is, to our knowledge, the first inter-laboratory variability budget for the TADF singlet–triplet gap, and it is the term that caps every predictor tested here. The quantity the model is regressed on, however, is the per-molecule *median*, whose precision is better than any single report: the RMS standard error of that median is 0.049 eV, about half the model MAE. Label imprecision is therefore a real term in the error budget but does not by itself account for it, and we do not claim the forest sits at a noise floor. Applying the noise-ceiling framework of Crusius *et al.*¹⁴ ($R_{\text{max}}^2 = 1 - \sigma_{\text{noise}}^2/\sigma_y^2$) quantifies this: the 0.049 eV median precision gives $R_{\text{max}}^2 \approx 0.91$ (95 % CI 0.85 to 0.94), and the single-report scatter (0.091 eV) gives $R_{\text{max}}^2 \approx 0.71$ (CI 0.48 to 0.81)—both well above the observed $R^2 = 0.26$. The model captures only 28–36 % of the noise-floor-limited ceiling depending on which floor applies, meaning label noise alone is not the dominant constraint. Progress requires both a better dataset *and* improvements in features or corpus scale; the data argument is justified because the label precision today (0.049 eV) is the term a curated benchmark would directly attack. A second, separate dispersion applies to computed references rather than to this

regression: the computed gap is functional-dependent. Across the 7 molecules (gas and toluene) for which we ran both B3LYP/6-31G(d) and CAM-B3LYP/def2-TZVP, ΔE_{ST} differs by a mean 0.30 eV (up to 0.58 eV) and changes relative magnitude (table S10); a higher-tier benchmark across six range-separated functionals on 27 emitters³² shows the same effect at finer resolution: a median inter-functional spread of 0.05 eV on a quantity that approximates *half* the gap, hence roughly 0.1 eV on ΔE_{ST} itself. Any claim of sub-0.1 eV accuracy against such a reference is not physically meaningful. This second effect bounds what a *computed* gap can mean; it does not bound a model regressed on measured labels, and we do not add the two into a single floor. Only the label term applies directly to the present task. What both establish is that richer feature sets and alternative regressors are not the missing ingredient. We tested both directly. No higher-capacity learner beats the random forest under the same scaffold CV. Histogram gradient boosting reaches 0.105 eV, an MLP 0.169 eV and an RBF SVR 0.120 eV (table S18); a second comparison over a differently configured set of learners places plain gradient boosting at 0.098 eV, within 0.002 eV of the forest and so inside the equivalence margin established above (`code/a007_leakage_and_nested_cv.py`; table S18 lists the histogram-boosting configuration). A frozen pretrained SMILES transformer (ChemBERTa, 77M parameters), the representation a current screening pipeline would reach for first, is no exception: its mean-pooled embeddings feed a random-forest or ridge head to MAE 0.109 eV to 0.113 eV on the same scaffold folds—it does not beat the 0.107 eV mean-value baseline, and is worse than Morgan by a paired 0.022 eV (95 % CI 0.008 to 0.034, excluding zero; `code/a013_pretrained_encoder_baseline.py`). Fine-tuning such an encoder on 231 labels would overfit and is not a meaningful test at this scale. Added model capacity therefore does not help, and no learned representation we tried improves on the hand-built descriptors, though we would not claim the forest is uniquely suited; and the negative lower R^2 bound is not a fingerprint-width artefact, since 512-, 1024- and 2048-bit Morgan models share it (table S18).

Stratifying by label quality does not isolate a label effect, and the attempt is instructive. Restricting to molecules that carry replicates, the model reaches MAE 0.074 eV on the 79 whose reports agree to ≤ 0.05 eV, 0.106 eV on the 23 whose reports do not, and 0.107 eV on the 129 for which label precision cannot be measured. That ordering is not evidence about labels: a constant predictor fitted within each stratum reproduces it almost exactly (0.074, 0.107 and 0.110 eV), because the strata differ in the dispersion of the target itself (standard deviation 0.10, 0.15 and 0.20 eV). Within any one stratum the model carries no advantage over a constant. We therefore report no stratified evidence for a label ceiling; the quantitative statement that survives is the precision of the median target, 0.049 eV (above).

A related question is whether *uncontrolled measurement conditions*, rather than label imprecision as such, set the limit. The corpus records no solvent, host or temperature for individual values, so a strictly condition-controlled subset cannot be built; the closest proxy is the source paper, since molecules whose values all come from one publication were measured under one laboratory’s conditions. Splitting the benchmark on that (218 single-source against 13 multi-source molecules) and scoring the random forest on the same scaffold cross-validation folds, the model’s advantage over a constant baseline predictor fitted within each group is 0.0071 and 0.0024 eV respectively, a difference of 0.005 eV with a 95 % CI of -0.014 to 0.035 that includes zero. The raw errors run the other way (0.058 against 0.093 eV) but the multi-source group’s target dispersion is half as large and its own constant predictor scores 0.061 eV, so that ordering is again dispersion rather than label quality. With only 13 molecules drawing on more than one paper the test constrains the effect rather than excluding it; we find no evidence that mixed conditions drive the error, and do not claim conditions are irrelevant.

The corpus size is not the limit either. Featurising 497 additional experimental emitters through the same pipeline (a 728-molecule set) raises the MAE from 0.098 to 0.124 eV, drives R^2 from 0.18 to -0.01 , and roughly halves the rank correlation (Spearman ρ from 0.34 to 0.16); the replicate-measured subset ($n = 465$) recovers only about half of the loss (table S19). Tripling the training data makes the prediction worse. The added molecules are resolved through a different structure pipeline, so wider chemical space and lower label quality are confounded and this test cannot separate them; what it does rule out is the reading that the corpus is simply too small.

A separate concern is the descriptor side: the NTO features are computed at the *vertical* (ground-state) geometry, whereas the measured gap reflects relaxed excited states. We bounded this approximation by recomputing ΔE_{ST} adiabatically for 14 representative emitters in gas and toluene at CAM-B3LYP/def2-TZVP with excited-state optimisation (table S12). Relative to these adiabatic references the vertical gap carries a mean absolute deviation of 0.195 eV, which exceeds the model’s own error. The deviation is dominated by an additive offset rather than by reordering—a linear fit gives slope 1.00 and intercept 0.07 eV ($R^2 = 0.70$, Spearman $\rho = 0.71$)—so relaxation shifts the scale more than it scrambles the ranking the triage relies on. The residual is not negligible, however: the RMS deviation is 0.245 eV and the worst molecule departs by 0.548 eV, with sign changes between compounds, and 14 molecules cannot settle how much of that is systematic. We take the vertical descriptors as an acceptable ranking input on this evidence, not as a validated approximation to the adiabatic gap.

3.5 Candidate prioritisation as a triage filter

The model’s practical value is as a coarse triage filter: it eliminates the bottom tier of candidates rather than ordering the viable subset. We quantify this directly. Taking emitters with experimental $\Delta E_{\text{ST}} \leq 0.10$ eV as the targets (49 % of the set), the structure-only model’s top-ranked candidates are enriched in targets relative to random selection by 1.2 to $1.4\times$ (precision rising from the 49 % base rate to 60 % at the top 10–20 and 68 % at the top 50); a bootstrap confidence interval on this cross-validation enrichment, given below, shows the smaller cutoffs are not statistically distinguishable from chance. Note that ΔE_{ST} is a necessary but not sufficient condition for TADF efficiency; emission wavelength matching is a subsequent, independent design step not addressed by this model. The filter therefore works, but weakly: it is a coarse first pass that modestly concentrates good emitters, not a selector. Part of that weakness is the threshold itself: 0.10 eV is only 0.03 eV above the 0.072 eV RMS inter-report scatter (section 3.4), so a molecule’s true class membership near that line is itself uncertain at the level of label noise—tightening the triage threshold runs directly into the same ceiling that bounds the regression task. Among well-characterised emitters, DMAC-DPS (experimental $\Delta E_{\text{ST}} \approx 0.09$ eV) and PXZ-NAI are examples of low-gap emitters the filter retains. Application to a specific lighting niche—horticultural or human-

centric—is a *separate, downstream* step: it requires matching the emission wavelength and oscillator strength, neither of which this scalar ΔE_{ST} model predicts. We therefore make no spectral-matching or device-level claim (reverse intersystem crossing rates and external quantum efficiencies depend on quantities this model does not resolve, and our earlier device estimates did not survive scrutiny, section 3.8).

3.6 A substructure-filtered extended chemical space

To indicate the chemical space such triage would operate over, we screened PubChem for structures carrying both a donor and an acceptor motif, retaining 22 194 (section 2.5; molecular weights 172 Da to 899 Da, 2 to 12 aromatic rings; property distributions in fig. S4). Every member passes the donor-acceptor pattern test, but the set carries the composition of the source database rather than of a designed emitter series, and its matches lean on the weaker motifs; it indicates scale, not a curated design space. We do not deposit a ranked candidate list. On the filtered set the split-conformal 90 % prediction interval spans 0.36 eV (mean half-width 0.18 eV), about 7.6 times the predicted gap, and for every one of the 100 lowest-predicted molecules the interval’s lower bound falls below zero. At this accuracy conformal prediction refuses selection at the single-molecule level; the filtered set only indicates the scale of the space that the labelled-data triage would operate over. The heuristic score of eq. (5) remains a planning aid, not a calibrated predictor, and the retained molecules are not quantum-validated.

Quantifying triage utility. On held-out labelled molecules the structure-only model enriches the low-gap class by 1.2 to 1.4 $\times$ over random selection, but a 95 % bootstrap confidence interval (5 000 resamples over molecules, the ranking re-derived from the fixed cross-validation out-of-fold predictions within each resample) shows this holds only at the less aggressive cutoffs. Against the 49 % base rate ($\Delta E_{\text{ST}} \leq 0.10$ eV), top-ranked precision reaches 60 % at both the top 10 and top 20 candidates (enrichment 1.22 $\times$, 95 % CI 0.63 to 1.89 and 0.80 to 1.70: both intervals include no enrichment), and 68 % at the top 50 (1.38 $\times$, CI 1.10 to 1.60) or 67.4 % at the top 20 % of the benchmark (1.37 $\times$, CI 1.08 to 1.61: both intervals exclude no enrichment). The enrichment is therefore only statistically established at cutoffs of roughly 50 molecules or a fifth of the benchmark; at the ten-to-twenty candidate scale a laboratory would actually shortlist, it cannot be distinguished from chance at this sample size. The full enrichment curve (fig. 2) peaks near 1.37 $\times$ at the top 5 to 20% and does not improve at more aggressive cuts; a split-conformal calibration lets a laboratory fix an operating point at a stated confidence. This is a statement about the labelled benchmark: the model supports *triage*, not selection, and the interval widths of section 3.6 rule out projecting a hit rate onto the unvalidated filtered library.

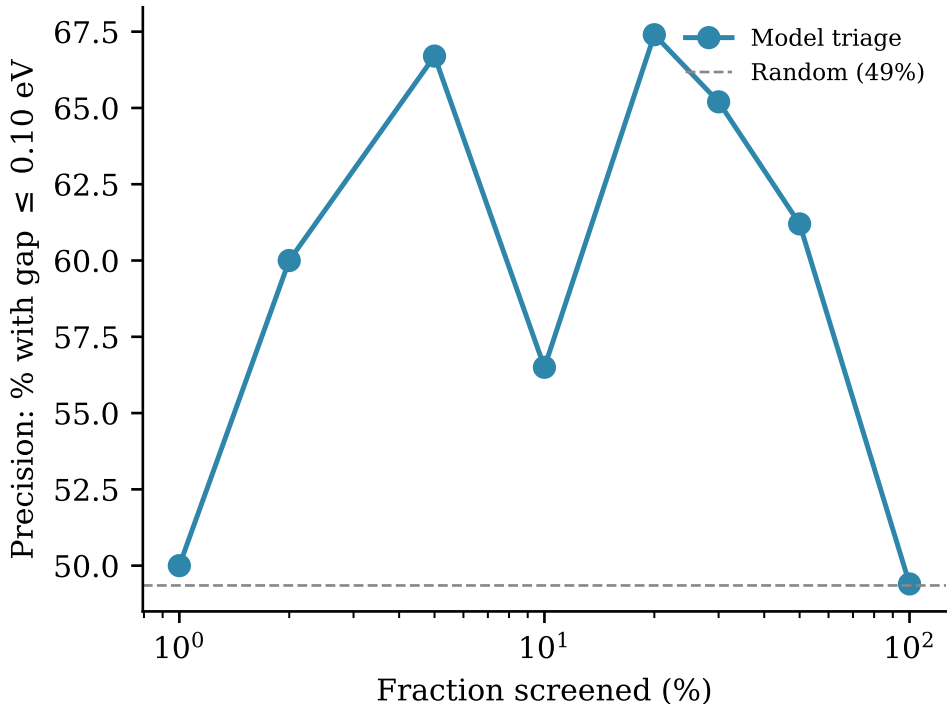


Figure 2 Triage precision versus fraction of the benchmark screened (log x). Dashed line: the 49 % random base rate. Enrichment is real but bounded at ≈ 1.3 – $1.4\times$; the top 1% is two molecules.

3.7 Comparison with existing ML approaches

Before comparing models, we contextualise the absolute errors against three baselines computed on the out-of-fold scaffold-CV predictions (table 2; 95 % bootstrap CIs, 2 000 resamples): the mean-value predictor (predicting $\bar{y}$ for every molecule, MAE = 0.107 eV), the median-value predictor (MAE = 0.098 eV), and a random-permutation null (mean MAE over 1 000 label shuffles, 0.120 eV, 95 % interval 0.113 to 0.127). The median is a tougher baseline than the mean because the experimental gap distribution

is right-skewed toward low-gap emitters; the permutation null is clearly worse than either constant predictor, confirming that the trained model exploits a real feature-label association rather than a fold artefact. Our best model (RF-Morgan, MAE = 0.091 eV) improves on the mean baseline by 15 %; the headline NTO model (MAE = 0.096 eV) improves by 10.6 % (0.011 eV, 95 % CI on the paired difference 0.003 to 0.022, excluding zero). The model edges even the skewed-median baseline (0.098 eV). This gain appears modest, and it should be read against the precision of the target: across the 102 multi-report molecules a single report deviates from the median by 0.072 eV on average (RMS), the worst molecule reaching a standard deviation of 0.56 eV, while the median actually regressed on carries an RMS standard error of 0.049 eV. An MAE of 0.096 eV is therefore about twice the precision of its own target — close enough that label quality is a material term, far enough that we do not describe the model as sitting at a floor. As shown above (section 3.3), the excited-state NTO descriptors offer no accuracy advantage over the purely structural sets, consistent with label noise dominating any feature advantage at the current scale.

Table 2 Naive baselines versus the structure-based model on the same 231-molecule out-of-fold scaffold-CV predictions. MAE in eV with 95 % percentile-bootstrap CI (2000 resamples); the permutation row is the mean and central 95 % of 1000 label shuffles. The median outperforms the mean (right-skewed target); the permutation null is worse than both, so the model captures real signal. Values from `code/compute_baselines.py`.

Predictor	MAE (eV)	95% CI
Mean predictor	0.107	(0.092, 0.124)
Median predictor	0.098	(0.081, 0.118)
Random-permutation null	0.120	(0.113, 0.127)
NTO-RF (this work)	0.096	(0.082, 0.109)

Prior ML for TADF gaps is best read by the *reference* it predicts, not by the headline error alone (table 3). The low MAEs in the literature are all against *computed* targets: Gómez-Bombarelli et al.⁷ trained a neural network to reproduce a TD-DFT gap inside a $>10^6$ -molecule screen, Thapa et al.³³ screened a large virtual library against semi-empirically computed gaps, and Barneschi et al.⁸ reach 0.02 eV with a 3D message-passing network—but against ADC(2) values, and for inverted-gap molecules rather than donor-acceptor TADF. Predicting the *experimental* gap is the harder task this paper addresses: Nigam et al.⁹ found direct regression of the measured gap so unreliable that they fell back to a good/bad classifier. Our 0.096 eV, under scaffold cross-validation and with label and reference noise accounted for, shows that cheap structural features nonetheless track the experimental gap to within ≈ 0.1 eV, about twice the precision of the target itself. The contribution is not a record accuracy but a calibrated, reproducible measure of how far cheap structural features reach, and a bounded demonstration that the excited-state step buys no accuracy on that task—while still buying rank order that transfers across source laboratories (section 3.2).

Table 3 Machine-learning prediction of TADF singlet-triplet gaps, ordered by reference type. Low MAEs are obtained only against *computed* gaps; against the *experimental* gap, errors are substantially larger and direct regression is sometimes abandoned. Only values reported in the cited sources are tabulated.

Study	Model (features)	Set (N)	Reference	MAE (eV)
Gómez-Bombarelli 2016 ⁷	NN surrogate	$>10^6$ screened	TD-DFT (computed)	reproduces TD-DFT ^a
Barneschi 2024 ⁸	3D message-passing GNN	INVEST set	ADC(2) (computed)	0.02 ^b
Nigam 2024 ⁹	DNN classifier (Morgan FP)	GA-evolved	ω B2PLYP' (computed)	regression poor ^c
This work	RF (NTO, structure)	231	experimental	0.096

^a Predicts the TD-DFT gap; no experimental-reference regression MAE reported. ^b Inverted-gap (Hund’s-rule-violating) molecules, not donor-acceptor TADF. ^c Direct gap regression reported as poor; the deployed model is a good/bad classifier (89 % to 98 % accuracy).

3.8 Limitations

This study is bounded in several ways. (1) The experimental corpus is heterogeneous (mixed solvents and methods, 0.072 eV RMS inter-report scatter); a single-source, solvent-controlled dataset would be required to push below the present ceiling. (2) The model predicts a consensus gap, not solvatochromic or conformer-dependent values. (3) We make no quantitative device claims. Molecule-independent estimates for spin-orbit coupling, k_{RISC} , and external quantum efficiency, derived from literature medians, were deliberately excluded from this analysis. (4) Ranking power is moderate ($\rho = 0.36$), so the model supports triage, not final candidate selection. (5) The target is a single scalar, $\Delta E_{\text{ST}}(S_1-T_1)$. A small S_1-T_1 gap is necessary but not sufficient for efficient delayed fluorescence: the splitting of the triplet manifold (T_1-T_2) also governs reverse intersystem crossing.³⁴ To gauge this we extracted T_1 and T_2 from the CAM-B3LYP TD-DFT calculations for 14 emitters (table S13): the T_1-T_2 gap ranges 0.02 eV to 1.50 eV and varies widely *among molecules with near-equal* ΔE_{ST} , so a scalar-gap triage cannot rank them on RISC propensity. The semi-empirical step is redundant for the S_1-T_1 gap (section 3.3) and does at least yield a triplet-manifold quantity that 2D structure does not, but we cannot show that quantity is reliable. Extending the sTDA extraction to all 231 emitters, 74 % carry a T_1-T_2 gap below 0.3 eV; that classification should be treated as indicative only, since the deviation of the sTDA manifold from the CAM-B3LYP reference (0.24 eV, below) is comparable to the 0.3 eV threshold being applied. We read this full-corpus manifold only qualitatively: benchmarked against the 14-molecule CAM-B3LYP reference (six molecules overlapping), the sTDA T_1-T_2 gap carries a mean absolute deviation of 0.24 eV and a rank correlation that is not resolved at this sample size (Spearman $\rho = 0.43$, $p = 0.40$, $n = 6$): the two methods agree on neither magnitude nor rank at a useful level, so quantitative ranking on RISC propensity

requires the higher-level calculation (table S13). Emitters with an *inverted* gap ($\Delta E_{\text{ST}} < 0$), which violate Hund’s rule through double-excitation character,³⁵ are out of scope, and the model fails on them exactly as its construction requires. A literature screen flags 5 corpus compounds for which at least one report gives an inverted gap; for two of these the consensus median we train on is itself negative (HAP-3MF, -0.22 eV; TPA-NZP, -0.038 eV), and these are the only 2 negative labels among the 231. The random forest—built on GFN2-xTB features that enforce Aufbau ordering—returns a positive gap for all five, so it carries the wrong sign for both molecules whose own training label is negative. The failure is representational rather than incidental. A regression forest can in principle return a negative value, since its output is an average over training labels and two of ours are negative; what it lacks is any feature that separates an inverted-gap molecule from a conventional one, because the sTDA descriptors are computed under Aufbau ordering and the structure-only descriptors carry no excited-state information at all. Predicting this class would require descriptors that encode double-excitation character, not more data (fig. S5). Screening this class would require ROKS- or EOM-CCSD-level descriptors.³¹ (6) The semi-empirical descriptors are computed at ground-state vertical geometries and therefore carry no information about excited-state structural relaxation, and none about spin-orbit coupling. Both matter physically for delayed fluorescence, and the omission is a plausible part of why the NTO features do not outperform 2D fingerprints: a vertical descriptor cannot represent the charge-transfer relaxation that separates two molecules with similar ground-state topology. Our 14-molecule adiabatic comparison bounds the size of the vertical approximation (mean absolute deviation 0.195 eV, RMS 0.245 eV) but cannot remove it, and excited-state optimisation for the full corpus was outside the compute budget. Descriptors built on relaxed excited-state geometries are the natural next test of whether the equivalence reported here survives. On the model side, the comparison covers tabular learners and one frozen pretrained molecular-transformer embedding (section 3.4); an end-to-end message-passing network or an encoder fine-tuned on a larger labelled set was not attempted, and either would need a benchmark larger than 231 molecules to be informative.

(7) Under a source-paper (DOI-grouped) split the model’s advantage over a mean-value predictor is not statistically resolved (section 3.2); demonstrated transfer is to new scaffolds measured under represented protocols, not to new measurement laboratories. Future work targets a curated experimental benchmark and uncertainty-aware models that report calibrated confidence alongside each prediction.

4 Conclusions

We measured how far cheap structure-based machine learning reaches against the experimental singlet–triplet gap ΔE_{ST} of TADF emitters, with every number reproducible from the deposited code. The study establishes where the field’s data, not its models, set the limit on predicting this quantity, and it isolates two effects that follow from that limit. Three findings stand out.

Structure alone predicts as well as the semi-empirical excited state. Random forests built from the 2D structure alone—Morgan fingerprints (MAE = 0.091 eV) and RDKit descriptors (0.100 eV)—predict experimental ΔE_{ST} as well as NTO descriptors derived from a semi-empirical excited-state calculation (0.096 eV), over 231 molecules and 212 scaffolds, and the result holds across scaffold, cluster and random cross-validation. The two representations agree to within 0.017 eV on paired folds, so the excited-state step buys no accuracy. The representations separate, however, on ranking under laboratory transfer: under a DOI-grouped split the NTO model retains Spearman $\rho = 0.35$ while Morgan drops to 0.16 ($\Delta\rho = 0.19$, 95 % CI 0.04 to 0.34, excluding zero only narrowly and on a single split). The excited-state features do not improve the value of the gap; they preserve the *order* of candidates across source laboratories—a dissociation between accuracy and ranking that, to our knowledge, has not been reported for this property, and one that a wider group-level interval could still absorb. SHAP on the NTO variant ties the signal to hole–electron overlap and charge-transfer character, keeping the physics-informed representation interpretable.

Data quality, not model capacity, is the binding term. The label bounds what any predictor can reach: independent reports of the same compound scatter by 0.072 eV RMS, and the per-molecule median actually regressed on is known only to 0.049 eV—about half the model error, so a real term in the budget but not the whole of it. Quantified across 231 molecules and 173 source papers, this is, to our knowledge, the first inter-laboratory variability budget for the TADF singlet–triplet gap. Error is lower on molecules whose replicate reports agree (0.074 eV against 0.106 eV), but that ordering is reproduced by a constant predictor and reflects the spread of the target within each stratum rather than label quality, so we draw no conclusion from it. At this data scale, capacity is not the missing ingredient: no higher-capacity learner we tested beats the random forest—a frozen pretrained molecular-transformer embedding reaches only MAE 0.109 eV to 0.113 eV—and a 609-core-hour range-separated TD-DFT reference gap does not either. Separately, the computed reference is itself functional-dependent (mean 0.30 eV, up to 0.58 eV between B3LYP and CAM-B3LYP); that bounds the meaning of a *computed* gap rather than the error of a model fitted to measured labels, and we do not combine the two into one floor. Claims of near-DFT precision against such references are not physically meaningful.

A general methodological caution for derived-energy targets. The leakage demonstration is not TADF-specific. Regressing ΔE_{ST} onto a feature set that contains the S_1 and T_1 excitation energies reproduces the target by construction ($\Delta E_{\text{ST}} \equiv E_{S_1} - E_{T_1}$); under cross-validation a linear model given those energies recovers the target exactly ($R^2 = 1.000$, MAE = 0) against $R^2 = 0.36$ without them, so the apparent accuracy of such models is arithmetic, not learning. Any target that is an algebraic combination of features already in the descriptor set will be “predicted” by arithmetic, invisibly to cross-validation. For a field in which many reported ML results target a computed gap, this is a reminder to benchmark against the measured quantity with features that exclude the constituent energies.

Practical scope. From the 2D structure alone, with no quantum chemistry, the model acts as a coarse triage filter—enriching low-gap candidates 1.2 to 1.4 $\times$ over random selection on held-out labelled data—useful for prioritisation but far from a selector, and not a quantitative photophysical method. Where ranking must survive a change of source laboratory the semi-empirical descriptors are the better choice, the one place in this study where the excited-state step pays for itself (section 3.2). On the filtered library

the conformal prediction intervals are too wide for molecule-level selection, so no ranked shortlist is deposited. Matching a specific lighting niche (horticultural or human-centric) needs the emission wavelength and oscillator strength, which this scalar-gap model does not predict, and is left to downstream work. We make no device-level performance claims: the spin-orbit-coupling-based k_{RISC} and external-quantum-efficiency estimates explored during this work were molecule-independent and have been withdrawn.

What would push the field past the ceiling. The 0.049 eV precision of the median label is the term a better dataset, not a better model, would attack. A curated, single-source, solvent-controlled experimental benchmark—one host, one measurement protocol, one laboratory per compound, along the lines the photophysics community is already converging on for how ΔE_{ST} should be reported¹³—would reduce the inter-report scatter that currently *is* the ceiling, and would let a retrained model and a stronger representation be distinguished from label noise. We do not run that benchmark here; we provide the quantitative evidence that it is the necessary next step. All data, trained models, and the scripts that regenerate every number and figure in this work are openly available (Zenodo DOI: 10.5281/zenodo.17436069), so that this benchmark can be extended and the limitations we report can be tested directly.

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Conflicts of interest

There are no conflicts to declare.

Author contributions

J.-P.T.N.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. S.C.T.K.: Software, Validation, Writing – review & editing. P.S.M.K.: Validation, Writing – review & editing. P.S.: Resources, Validation, Writing – review & editing. S.G.N.E.: Supervision, Project administration, Writing – review & editing.

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Declaration of generative AI and AI-assisted technologies

In preparing this work the authors used AI assistants (Claude by Anthropic — Sonnet 4.5, accessed May 2026; Sonnet 5 and Opus 5, accessed September 2026 — and ChatGPT by OpenAI). Their use covered literature triage, L^AT_EX formatting and consistency checking, drafting and revision of manuscript text, and writing the analysis scripts whose outputs are reported here — among them the paired equivalence and nested cross-validation tests, the enrichment and label-precision audits, and the pretrained-encoder baseline. The deposited analysis ledger records the provenance of every value. The AI assistants are not authors and made no independent scientific judgement; all numerical values, computational methods and interpretations were checked against the deposited data and approved by the authors, who take full responsibility for them.

Data availability

All experimental data, descriptor tables, trained models, and analysis code are openly available on Zenodo under DOI: 10.5281/zenodo.17436069 (CC-BY-4.0 license). The deposit includes: (1) the experimental ΔE_{ST} corpus and the 231-molecule benchmark set (212 scaffolds); (2) the 35 NTO spatial descriptors and the structure-only Morgan and RDKit feature tables; (3) the DFT functional benchmark and the CAM-B3LYP vertical-versus-adiabatic and triplet-manifold data (14 emitters); (4) the trained random-forest models, SHAP attributions, and validation outputs (scaffold and Butina-cluster cross-validation, label-permutation test, capacity tests); (5) the heuristic 22 194-molecule library with the conformal calibration script and its interval widths—no ranked candidate list is deposited, because at this accuracy the intervals do not support molecule-level selection; and (6) all analysis and figure-generation scripts that regenerate every reported number.

Notes and references

- [1] H. Uoyama, K. Goushi, K. Shizu, H. Nomura and C. Adachi, *Nature*, 2012, **492**, 234–238.
- [2] M. Yokoyama, K. Inada, Y. Tsuchiya, H. Nakanotani and C. Adachi, *Chemical Communications*, 2018, **54**, 8261–8264.
- [3] B. Bugbee, *Acta Horticulturae*, 2016, **1134**, 1–12.
- [4] S. Curtarolo, G. L. W. Hart, M. B. Nardelli, N. Mingo, S. Sanvito and O. Levy, *Nature Materials*, 2013, **12**, 191–201.
- [5] K. T. Butler, D. W. Davies, H. Cartwright, O. Isayev and A. Walsh, *Nature*, 2018, **559**, 547–555.
- [6] E. O. Pyzer-Knapp, J. W. Pitera, P. W. J. Staar, S. Takeda, T. Laino, D. P. Sanders, J. Sexton, J. R. Smith and A. Curioni, *npj Computational Materials*, 2022, **8**, 84.
- [7] R. Gómez-Bombarelli, J. Aguilera-Iparraguirre, T. D. Hirzel, D. Duvenaud, D. Maclaurin, M. A. Blood-Forsythe, H. S. Chae, M. Einzinger, D.-G. Ha, T. Wu, G. Markopoulos, S. Jeon, H. Kang, H. Miyazaki, M. Numata, S. Kim, W. Huang, S. I. Hong, M. Baldo, R. P. Adams and A. Aspuru-Guzik, *Nature Materials*, 2016, **15**, 1120–1127.

- [8] L. Barneschi, L. Rotondi and D. Padula, *J. Phys. Chem. A*, 2024, **128**, 2417–2426.
- [9] A. Nigam, R. Pollice, P. Friederich and A. Aspuru-Guzik, *Chem. Sci.*, 2024, **15**, 2618–2639.
- [10] J.-P. Tchapet Njafa, E. V. Kameni Tcheuffa, A. Maghame Fomkpo and S. G. Nana Engo, *Journal of Chemical Information and Modeling*, 2026, **66**, 2230–2248.
- [11] J.-P. Tchapet Njafa, E. V. Kameni Tcheuffa, A. M. Fomkpo and S. G. Nana Engo, *Journal of Chemical Information and Modeling*, 2026.
- [12] B. Das and A. Mondal, *J. Mater. Chem. A*, 2025, **13**, 33553–33570.
- [13] A. Shuaib, L. Salah, A. Prlj, N. Rajendran, M. K. Etherington, A. A. Nazeer, C. S. Ponseca, A. P. Monkman, A. Danos and S. Makhseed, *J. Mater. Chem. C*, 2026, **14**, 4353–4361.
- [14] D. Crusius, F. Cipcigan and P. C. Biggin, *Faraday Discuss.*, 2025, **256**, 304–321.
- [15] E. Heid, C. J. McGill, F. H. Vermeire and W. H. Green, *J. Chem. Inf. Model.*, 2023, **63**, 4012–4029.
- [16] S. Kaufman, S. Rosset, C. Perlich and O. Stitelman, *ACM Trans. Knowl. Discov. Data*, 2012, **6**, 1–21.
- [17] S. Kapoor and A. Narayanan, *Patterns*, 2023, **4**, 100804.
- [18] N. Artrith, K. T. Butler, F.-X. Coudert, S. Han, O. Isayev, A. Jain and A. Walsh, *Nature Chemistry*, 2021, **13**, 505–508.
- [19] G. W. Bemis and M. A. Murcko, *Journal of Medicinal Chemistry*, 1996, **39**, 2887–2893.
- [20] Y. Olivier, J.-C. Sancho-García, L. Muccioli, G. D’Avino and D. Beljonne, *The Journal of Physical Chemistry Letters*, 2018, **9**, 6149–6163.
- [21] C. Bannwarth, S. Ehlert and S. Grimme, *Journal of Chemical Theory and Computation*, 2019, **15**, 1652–1671.
- [22] S. Grimme and C. Bannwarth, *The Journal of Chemical Physics*, 2017, **145**, 074108.
- [23] R. L. Martin, *The Journal of Chemical Physics*, 2003, **118**, 4775–4777.
- [24] G. Landrum *et al.*, *RDKit: Open-source cheminformatics*, <https://www.rdkit.org>, 2016, Version 2026.03.
- [25] L. Breiman, *Machine Learning*, 2001, **45**, 5–32.
- [26] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot and E. Duchesnay, *Journal of Machine Learning Research*, 2011, **12**, 2825–2830.
- [27] F. Neese, F. Wennmohs, U. Becker and C. Riplinger, *The Journal of Chemical Physics*, 2020, **152**, 224108.
- [28] S. M. Lundberg and S.-I. Lee, *Advances in Neural Information Processing Systems*, 2017, **30**, 4765–4774.
- [29] Y. Liu, M. Nishiura, Y. Wang and Z. Hou, *Journal of the American Chemical Society*, 2012, **134**, 5592–5593.
- [30] M. Y. Wong and E. Zysman-Colman, *Advanced Materials*, 2017, **29**, 1703554.
- [31] E. Salvi, G. Agostini, S. Veglianti, G. J. Costa, L. De Vico, D. Padula and C. A. Guido, *J. Comput. Chem.*, 2025, **46**, e70267.
- [32] T. Froitzheim, L. Kunze, S. Grimme, J. M. Herbert and J.-M. Mewes, *J. Phys. Chem. A*, 2024, **128**, 6324–6335.
- [33] K. Thapa, J. I. Jones and L. E. Ratcliff, *J. Mater. Chem. C*, 2025, **13**, 19212–19225.
- [34] R. Keruckiene, A. A. Vaitusionak, M. I. Hulnik, I. A. Berezianko, D. Gudeika, S. Macionis, M. Mahmoudi, D. Volyniuk, D. Valverde, Y. Olivier, K. L. Woon, S. V. Kostjuk, S. Reineke, J. V. Grazulevicius and G. Sini, *Journal of Materials Chemistry C*, 2024, **12**, 3450–3464.
- [35] S. Veglianti, A. Michieletti and D. Padula, *ACS Materials Letters*, 2025, **7**, 3220–3226.